Project Idea 2: Supply Chain Dataset Analysis

Project Implementation Plan & Deliverables Breakdown (4 Weeks)

Project Overview: This project focuses on performing an end-to-end analysis of a supply chain dataset using **Power BI**. The objective is to clean raw data, formulate decision-oriented analytical questions, build an interactive dashboard, and present actionable business insights to organizational stakeholders.

Week 1: Data Cleaning and Preprocessing

Tools: Power BI

- **Tasks:**
  - **Data Preprocessing:** Clean and preprocess the dataset using Power BI (handling missing values, removing duplicates, and transforming data types).
  - **Data Modeling:** Establish table relationships and setup initial DAX calculated columns and measures.
- **Deliverables:**
  - Cleaned dataset ready for analysis.
  - Data preprocessing notebook / transformation log.

Week 2: Analysis Questions Phase

Tools: Power BI

- **Tasks:**
  - **Determine Data Analysis Questions:** Determine all possible analysis questions that can be deduced from the given dataset and would be of interest to the organization's decision makers (e.g., impact of product category on revenue).
- **Deliverables:**
  - Set of analysis questions that can be answered via the dataset.

Focus Area	Sample Analytical Questions
Revenue & Products	What is the impact of product category on overall revenue and profit margins?
Logistics & Shipping	What are the average shipping lead times and freight costs across different geographic regions?
Inventory Management	Which product lines face the highest risk of stockouts or excessive holding costs?

Week 3: Dashboard Phase

Tools: Power BI

- **Tasks:**
  - **Build Dashboard:** Build a Power BI dashboard that visualizes the answers to the asked questions.
  - **Visual Design:** Configure appropriate charts (KPI Cards, Bar Charts, Trend Analysis, Maps) for clear communication.
  - **Interactivity:** Implement slicers, filters, and drill-down features for interactive exploration.
- **Deliverables:**
  - Interactive Power BI dashboard (`.pbix` file).

Week 4: Final Presentation

Tools: Power BI, PowerPoint / Report

- **Tasks:**
  - **Final Presentation:** Prepare a report and presentation summarizing the project work, including data analysis, model development, and insights deployment.
  - **Business Insights:** Compile key findings and strategic recommendations for decision makers.
- **Deliverables:**
  - Final report and presentation slides.

Project Roadmap Summary

Week	Phase	Primary Tools	Key Deliverables
Week 1	Data Cleaning & Preprocessing	Power BI (Power Query)	Cleaned Dataset + Preprocessing Log
Week 2	Analysis Questions Phase	Power BI / Business Analysis	Set of Analysis Questions
Week 3	Dashboard Phase	Power BI Desktop	Power BI Interactive Dashboard
Week 4	Final Presentation	Power BI, PowerPoint / PDF	Final Report & Presentation